🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.AT.8

Understand Equations and Their Solutions

Lesson 8-1 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

COLD CASE MISSION

The Equation Vault

You are Detective Ruiz's new code analyst. A tip just came in: 'A number of stolen gems plus 8 more equals 20 total.' The evidence vault only opens when the clue is translated into a correct equation. Master writing equations and the case cracks wide open.

My Notes Level 2 Standard

📖 **TODAY'S OBJECTIVES**

Content Objective: I can write an equation for a real situation and check whether a given value is a solution by substituting it.
Language Objective: I can explain a solution using the words substitute, solution, true, and equal sign.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Write Equations | Equation | Variable | Equal sign | Expression | Constant | Unknown | Solution | Substitute

Tap any word to see what it means and a picture.

1. Two expressions joined by an __________ __________ make an equation.

2. A math sentence that says two amounts are equal, with an = sign, is an __________.

3. A letter that stands for an unknown number is a __________.

4. The symbol = that shows two amounts are the same is the __________.

5. A group of numbers and variables with no equal sign is an __________.

6. A fixed number that does not change is a ______.

7. The value you are trying to find in an equation is the ______.

8. A value that makes the equation true when ______ for the variable is its solution.

9. Replace the variable with a number to ______ whether it is a solution.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How many paintings has the detective recovered? How did writing an equation help you understand the situation?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Write Equations** — Represent a relationship with expressions joined by an equal sign.
  Escribir ecuaciones — Representar una relación con expresiones unidas por un signo igual.
- [ ] **Equation** — A math sentence with an equal sign showing both sides are the same.
  Ecuación — Una oración matemática con un signo igual que muestra que ambos lados son iguales.
- [ ] **Variable** — A letter that stands for a number that is unknown or can change.
  Variable — Una letra que representa un número desconocido o que puede cambiar.
- [ ] **Equal sign** — The = sign, showing both sides are the same.
  Signo igual — El signo =, que muestra que ambos lados son iguales.
- [ ] **Expression** — A math phrase with numbers and letters, but no equal sign.
  Expresión — Una frase matemática con números y letras, pero sin signo igual.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The detective recovered ____ paintings because p + 14 = 37 means p = 37 − ____ = ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: I let a variable stand for the unknown number of gems.

Evidence: Students choose a variable (such as n) for the unknown gems, build n + 8 = 20, and connect 'plus 8 more' to + 8 and 'equals 20 total' to = 20.

Reasoning: This evidence connects the definition of equation to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective recovered ____ paintings because p + 14 = 37 means p = 37 − ____ = ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.
Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Write Equations
2. **Notes 2:** Equation
3. **Notes 3:** Variable
4. **Notes 4:** Equal sign
5. **Notes 5:** Expression
6. **Notes 6:** Constant
7. **Notes 7:** Unknown
8. **Notes 8:** Solution
9. **Notes 9:** Substitute
10. **Watch for this mistake:** *A common mistake in Write Equations is writing subtraction and division clues in the wrong order — putting the number from the clue first instead of the variable first. For 'a number decreased by 8 equals 5,' students often write $8 - n = 5$ instead of the correct $n - 8 = 5$; for 'a number divided by 4 equals 6,' they write $4 \div n$ (or $4n$) instead of $n / 4 = 6$. Always start with the variable, then attach the operation in the order the words describe it, and check by substituting your answer back in.*
11. **Try It 1:** A. $5b = 35$ — *'5 times the number' means 5 times b, so $5b = 35$. Check: $5 \times 7 = 35$.*
12. **Try It 2:** A. $m - 9 = 24$ — *'9 fewer' means subtract 9, so $m - 9 = 24$. Check: $33 - 9 = 24$.*